

Structured Positive-Kernel Elimination and Gramian Rigidity as a Reusable Toolkit

Research note

Abstract

Assuming that the candidate proof of the scalar Crouzeix conjecture survives expert review, its central mechanism can be separated into a reusable two-stage method: structured elimination of nuisance terms from a positive kernel, followed by a rigidity argument for two weighted Gramians. The method is not valid for arbitrary positive kernels. Its essential hypothesis is an algebraic alignment that makes the nuisance module exactly annihilable.

1 The guiding abstraction

The proposed toolkit may be summarized as

$$\boxed{\text{structured positive-kernel elimination} \implies \text{Gramian rigidity.}} \quad (1)$$

The first stage removes an unknown but structured term without an estimate. The second stage converts the surviving positive-matrix inequality into a sharp norm bound through detectability and a Stein identity.

2 A kernel-elimination principle

Suppose that

$$H(z) = F(z) + E(z), \quad \operatorname{Re} H(z) \succeq 0, \quad (2)$$

where F is known and the nuisance term E is constrained to a linear space $\mathcal{E}$. Positivity of the real part produces the Herglotz kernel

$$L_H(z, w) = \frac{H(z) + H(w)^*}{1 - z\bar{w}} \succeq 0. \quad (3)$$

Choose points $z_1, \dots, z_N \in \mathbb{D}$ and vectors $\xi_1, \dots, \xi_N$, and set

$$\eta_i = \sum_{j=1}^N \frac{\xi_j}{1 - z_i \bar{z}_j}. \quad (4)$$

The contribution of E to the sampled kernel quadratic form is exactly

$$2 \operatorname{Re} \sum_{i=1}^N \xi_i^* E(z_i) \eta_i. \quad (5)$$

Consequently, test data satisfying

$$\operatorname{Re} \sum_{i=1}^N \xi_i^* E(z_i) \eta_i = 0 \quad (E \in \mathcal{E}) \quad (6)$$

remove the entire nuisance module from the positive-kernel inequality. What remains depends only on the known part F .

In the candidate Crouzeix proof, after an adjoint similarity the nuisance module has the form

$$E(z) = D(z)G, \quad D(z) \text{ diagonal}, \quad (7)$$

and the auxiliary vector v is determined by the single equation

$$Gv + Pu = 0. \quad (8)$$

This equation constructs an annihilator for every diagonal analytic correction simultaneously. The step may therefore be regarded as an *annihilator compression* of a positive kernel.

3 A standalone Gramian-rigidity lemma

The second part of the mechanism is independent of numerical ranges.

Lemma 3.1 (Detector–anticommutator rigidity). *Let $P = P^*$ and $X = X^*$ be finite matrices satisfying*

$$P \succeq I, \quad X \succeq 0, \quad cX - XP - PX \succeq 0, \quad c \geq 2. \quad (9)$$

Assume the detectability condition

$$\ker X \subseteq \ker(P - I). \quad (10)$$

Then

$$P \preceq \frac{c}{2}I. \quad (11)$$

If, in addition,

$$P = I + aC^*PC, \quad a > 0, \quad (12)$$

then

$$\|C\|^2 \leq \frac{c/2 - 1}{a}. \quad (13)$$

Proof. Let $Pe = pe$ for a unit eigenvector e . Testing the third inequality in (9) on e gives

$$0 \leq (c - 2p)e^*Xe. \quad (14)$$

If $p > c/2$, positivity of X forces $e^*Xe = 0$, and hence $Xe = 0$. Detectability then gives $Pe = e$, contradicting $p > c/2 \geq 1$. Thus every eigenvalue of P is at most $c/2$, which proves (11).

Finally, $P \succeq I$ and (12) imply

$$aC^*C \preceq aC^*PC = P - I \preceq \left(\frac{c}{2} - 1\right)I.$$

This is (13). □

The same detector argument extends to a completely positive dynamic

$$\Phi(Y) = \sum_j C_j^* Y C_j. \quad (15)$$

Whenever the corresponding Stein equation and detectability condition hold, it gives bounds on the row energy $\sum_j C_j^* C_j$. This makes Lemma 3.1 potentially useful in multivariable realization and control problems.

4 Scale selection and the distinguished radius $1/2$

The two scales in the candidate proof arise from an optimization principle. For $0 < r < 1$, sampling at

$$z_i = r\overline{\lambda_i} \quad (16)$$

produces the weighted Gramians

$$P_r = \sum_{k \geq 0} r^{2k} C^{*k} C^k, \quad (17)$$

$$R_r = \sum_{k \geq 0} r^k C^{*k} C^k, \quad (18)$$

$$X_r = R_r - P_r = \sum_{k \geq 1} (r^k - r^{2k}) C^{*k} C^k \succeq 0. \quad (19)$$

Exact cancellation yields

$$\frac{2}{1-r} X_r - X_r P_r - P_r X_r \succeq 0. \quad (20)$$

The detector condition is automatic here: $X_r e = 0$ implies $Ce = 0$, and then the series for P_r gives $P_r e = e$. Lemma 3.1, with $c = 2/(1-r)$, therefore gives

$$P_r \preceq \frac{1}{1-r} I. \quad (21)$$

The Stein identity

$$P_r = I + r^2 C^* P_r C \quad (22)$$

then yields

$$\|C\| \leq \frac{1}{\sqrt{r(1-r)}}. \quad (23)$$

Since $r(1-r)$ is uniquely maximized at $r = 1/2$, the optimal choice is

$$r = \frac{1}{2}, \quad \|C\| \leq 2. \quad (24)$$

Thus $1/2$ is not merely a fortunate sampling value: it is the unique optimizer of the two-scale family.

5 Immediate structural extensions

The simple-spectrum assumption is not the essential feature of the cancellation. Suppose that

$$T = S \Lambda S^{-1} \quad (25)$$

is diagonalizable, possibly with repeated eigenvalues, and that the correction belongs after adjoint similarity to a row-separable diagonal module

$$\mathcal{E}_S = \{S^{-*} D S^* : D \text{ diagonal}\}. \quad (26)$$

The same rowwise annihilator construction applies. If the correction belongs specifically to $\text{alg}(T^*)$, repeated eigenvalues merely impose additional equalities among the relevant diagonal entries.

This observation points toward extensions to finite-dimensional semisimple commutative algebras and simultaneously diagonalizable commuting tuples. Removing diagonalizability would require an additional argument, such as radial regularization followed by a controlled perturbation; this is a proposed direction rather than a consequence established here.

6 Promising open-problem targets

6.1 The matrix-valued Crouzeix problem

The double-layer map is completely positive, so its amplification to matrix-valued functions preserves positivity without loss. The new obstruction is algebraic: scalar diagonal nuisance terms become noncommuting matrix blocks. A block, bimodule, or tangential version of the annihilator construction would therefore be a major advance. For the complete spectral-set formulation, see the paper of Davidson, Paulsen, and Woerdeman, [arXiv:1612.05683](#).

6.2 The abstract Crouzeix conjecture

Clouâtre, Ostermann, and Ransford conjecture that additional unital antilinear structure should force the constant 2; see [arXiv:2011.10422](#). The present mechanism establishes this type of conclusion when the adjoint correction lies in a semisimple annihilable module. A broader criterion guaranteeing an annihilator could connect the two methods.

6.3 Scaled q -numerical ranges

Recent work extends the Crouzeix–Palencia framework to scaled q -numerical ranges and proposes a corresponding Crouzeix-type conjecture; see O’Loughlin–Rani, [arXiv:2603.15536](#). The associated kernel has a comparatively simple nuisance term, so one can search for q -dependent pairs of Gramian weights and then optimize their scale as in (24).

6.4 Annular and multiply connected spectral constants

Annulus theory combines double-layer positivity with coupled information about T and T^{-1} ; see Jury–Tsikalas, [arXiv:2307.13387](#). Current work on quantum-annulus constants obtains asymptotically sharp bounds; see Pascoe, [arXiv:2505.06230](#). This setting naturally suggests paired forward and backward Gramians, together with more than one cancellation anchor.

6.5 Multivariable Schur–Agler problems

Positive-kernel descriptions of the Schur–Agler norm involve diagonalizable tuples and semidefinite optimization; see Hartz–Wang, [arXiv:2602.14047](#). Kernel annihilation might eliminate nonunique Agler components before a multi-index Gramian argument is applied. This direction is more speculative because several positive kernels and several Stein recursions occur simultaneously.

7 Why adjoint alignment is indispensable

The orientation of the nuisance algebra cannot be discarded. For $M > 0$, let

$$T_M = \begin{pmatrix} 0 & M \\ 0 & 1/2 \end{pmatrix}, \quad H(z) \equiv I. \quad (27)$$

Then

$$\sigma(T_M) \subset \mathbb{D}, \quad \operatorname{Re} H = I, \quad (28)$$

and

$$H(z) - (I - zT_M)^{-1} = - \sum_{k \geq 1} z^k T_M^k \in \operatorname{alg}(T_M). \quad (29)$$

Nevertheless, $\|T_M\| \rightarrow \infty$ as $M \rightarrow \infty$. Thus replacing $\text{alg}(T^*)$ by $\text{alg}(T)$ destroys the conclusion even though both algebras are commutative. The adjoint orientation is exactly what makes the nuisance term compatible with the annihilating test data.

8 Research program

The reusable content of the mechanism is therefore the following five-step program:

1. Find a positive kernel.
2. Identify an adjoint-aligned nuisance module.
3. Construct test vectors that annihilate the entire module.
4. Arrange the surviving terms as an ordered difference of weighted Gramians.
5. Verify detectability and a Stein recursion.

The method is credible precisely where all five steps can be carried out without estimating the nuisance term. The complete Crouzeix problem and scaled q -numerical-range problems are the most direct targets; annular and multivariable variants require additional forward–backward or multi-index structure.